Market { Firms : max profit . constraint : cost
HHS : max happiness. constraint : budget

Students:
max mark
constraint : time
{ sleep
Fun

ECON1050 Tutorial 6

Single Variable Optimization

Week 8 - Semester 1 2026

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Today's Plan

- Recap: key ideas from Lecture 6
- Tutorial questions: Q1–Q5

Recap: Optimization and global extreme points

Optimization

Optimization means finding the best outcome of a function over its domain, such as the highest profit or the lowest cost.

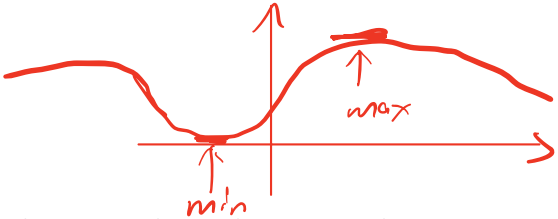
Global extreme points

For a function f with domain D:

$$c \text{ is a maximum point if } f(x) \leq f(c) \text{ for all } x \in D,$$

$$c \text{ is a minimum point if } f(x) \geq f(c) \text{ for all } x \in D.$$

A maximum or minimum point over the whole domain is a **global extreme point**.



Recap: Stationary points and the first-order condition

Stationary points

If f is differentiable and has a maximum or minimum at an interior point c, then the tangent line is horizontal there:

$$f'(c) = 0.$$

Points where $f'(c) = 0$ are called **stationary points** (or critical points).

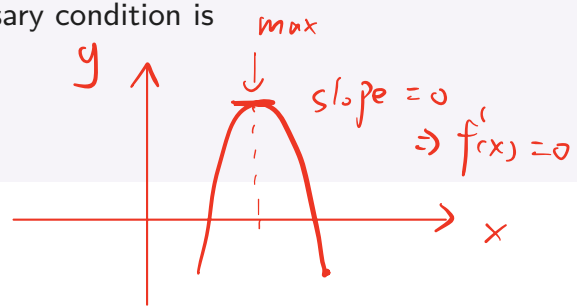
First-order condition (FOC)

For an interior optimum of a differentiable function, a necessary condition is

$$f'(x) = 0.$$

This condition is necessary, but not sufficient.

point is optimum point $\Rightarrow f'(x) = 0$



Recap: Finding global extrema on a closed interval

Extreme Value Theorem

If f is continuous on a closed and bounded interval $[a, b]$, then f attains both a maximum and a minimum on $[a, b]$.

$[a, b]$

Closed-interval method

To find global extrema of a differentiable function on $[a, b]$:

1. Find all stationary points in (a, b). $f'(x) = 0$
2. Evaluate f at all stationary points and at the endpoints a and b.
3. Compare the values.

Important caution

If the interval is not closed or not bounded, a maximum or minimum need not exist.

Recap: Local extrema and the first derivative test

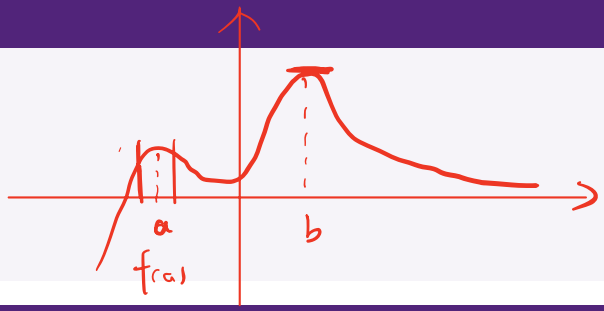
Local extreme points

A function has a **local maximum** (minimum) at c if $f(c)$ is the largest (smallest) value in some neighbourhood of c.

Candidates for extreme points

Possible global or local extrema can occur at:

- stationary points where $f'(x) = 0$,
- endpoints of the domain or interval,
- points where f is not differentiable.



First derivative test

- If f' changes from positive to negative, then f has a local maximum.
- If f' changes from negative to positive, then f has a local minimum.
- If the sign of f' does not change, then there is no local extreme point.

$f'(x)$

Recap: Second derivative test, concavity, and convexity

Second derivative test

If $f'(c) = 0$ and f is twice differentiable, then:

- $f''(c) < 0$ implies a local maximum at c,
- $f''(c) > 0$ implies a local minimum at c,
- $f''(c) = 0$ is inconclusive.

Concavity and convexity

- $f''(x) < 0$ on an interval implies f is **strictly concave** there.
- $f''(x) > 0$ on an interval implies f is **strictly convex** there.

If a concave (convex) function has a stationary point in the interior of an interval, then that point is a maximum (minimum).

Recap: Inflection points

Definition

A point c is an **inflection point** if the curvature changes there, that is, if f'' changes sign from positive to negative or from negative to positive.

Test for inflection points

- If c is an interior inflection point and f'' is continuous, then $f''(c) = 0$.
- If $f''(c) = 0$ and f'' changes sign at c, then c is an inflection point.

Third derivative hint

test

If $f''(c) = 0$ and $f'''(c) \neq 0$, then c is an inflection point.

Tutorial Q1

Question 1

Find the maximum and minimum values of each function on the stated interval.

(a) $f(x) = x^3 - 3x + 8, \quad x \in [-1, 2]$

(b) $f(x) = x^5 - 5x^3, \quad x \in [-1, \sqrt{5}]$

Closed-interval method

To find global extrema of a differentiable function on $[a, b]$:

1. Find all stationary points in (a, b). $f'(x) = 0$
2. Evaluate f at all stationary points and at the endpoints a and b.
3. Compare the values.

(a) $f(x) = x^3 - 3x + 8$ on $[-1, 2]$

$f'(x) = 3x^2 - 3 = 3(x+1)(x-1) = 0$

$\Rightarrow x = -1, 1$ stationary points

$f(-1) = (-1)^3 - 3(-1) + 8 = -1 + 3 + 8 = 10$

$f(1) = 1 - 3 + 8 = 6$

$f(2) = 8 - 6 + 8 = 10$

Global max: $f(-1)$ and $f(2)$, 10

Global min: $f(1) = 6$

(b) $f(x) = x^5 - 5x^3$ on $\boxed{[-1, \sqrt{5}]}$

$f'(x) = 5x^4 - 15x^2 = 5x^2(x^2 - 3) = 0$

$\Rightarrow x = 0, \sqrt{3}, \cancel{-\sqrt{3}}$ stationary points

$f(0) =$

$f(\sqrt{3}) =$

$f(-1) =$

$f(\sqrt{5}) =$

$\Rightarrow$ compare them

Tutorial Q1(a): Global extrema on a closed interval

Knowledge used

- Use the Extreme Value Theorem to guarantee existence on a closed interval.
- Find stationary points in the interior.
- Compare values at stationary points and endpoints.

Solution

For $f(x) = x^3 - 3x + 8$ with $x \in [-1, 2]$, we first find stationary points:

$$f'(x) = 3x^2 - 3 = 3(x^2 - 1).$$

Thus $f'(x) = 0$ at $x = -1$ and $x = 1$, but only $x = 1$ lies in $(-1, 2)$ with $f(1) = 1 - 3 + 8 = 6$. Now evaluate the endpoints:

$$f(-1) = 10, \qquad f(2) = 10.$$

Hence the global minimum is 6 at $x = 1$, and the global maximum is 10 at both $x = -1$ and $x = 2$.

Tutorial Q1(b): Another closed-interval problem

Knowledge used

- Solve $f'(x) = 0$ to find interior stationary points.
- Check which stationary points lie in the interval.
- Compare all candidate values.

Solution

For $f(x) = x^5 - 5x^3$ with $x \in [-1, \sqrt{5}]$, we have

$$f'(x) = 5x^4 - 15x^2 = 5x^2(x^2 - 3).$$

So $f'(x) = 0$ at $x = -\sqrt{3}, 0, \sqrt{3}$. Only $x = 0$ and $x = \sqrt{3}$ lie in $(-1, \sqrt{5})$.

$$f(0) = 0, \qquad f(\sqrt{3}) = (\sqrt{3})^5 - 5(\sqrt{3})^3 = -6\sqrt{3}.$$

At the endpoints, $f(-1) = 4$ and $f(\sqrt{5}) = 0$. Hence the global maximum is 4 at $x = -1$, and the global minimum is $-6\sqrt{3}$ at $x = \sqrt{3}$.

Tutorial Q2

Question 2

Determine the stationary points of

$f'(x)$

$\mathbb{R}$ is closed but not bounded

$$f(x) = x^3 + 3x^2 - 2,$$

and classify them using the second derivative test. $f''(x)$

Second derivative test

If $f'(c) = 0$ and f is twice differentiable, then:

- $f''(c) < 0$ implies a local maximum at c,
- $f''(c) > 0$ implies a local minimum at c,
- $f''(c) = 0$ is inconclusive.

$f(x) = x^3 + 3x^2 - 2$

$f'(x) = 3x^2 + 6x$

$f''(x) = 6x + 6$

Stationary: $f'(x) = 3x^2 + 6x = 0$

$= 3x(x+2) = 0$

$\Rightarrow x = 0, -2$

$f''(0) = 6 > 0 \Rightarrow$ local min

$f''(-2) = 6 \cdot (-2) + 6 = -6 < 0 \Rightarrow$ local max

Tutorial Q2: Classifying stationary points

Knowledge used

- Use the FOC $f'(x) = 0$ to find stationary points.
- Apply the second derivative test to classify each point.

Solution

$$f'(x) = 3x^2 + 6x = 3x(x+2).$$

So the stationary points are at

$$x = -2 \quad \text{and} \quad x = 0.$$

Next,

$$f''(x) = 6x + 6.$$

Therefore,

$$f''(-2) = -6 < 0 \Rightarrow \text{local maximum at } x = -2,$$

$$f''(0) = 6 > 0 \Rightarrow \text{local minimum at } x = 0.$$

Tutorial Q3

Question 3

For

Stationary point $f'(x)$

$$f(x) = x^3 e^{-x},$$

find the possible extreme points and inflection points.

Test for inflection points

- If c is an interior inflection point and f'' is continuous, then $f''(c) = 0$.
- If $f''(c) = 0$ and f'' changes sign at c, then c is an inflection point.

$f(x) = x^3 e^{-x}$ product rule

$f'(x) = 3x^2 \cdot e^{-x} + x^3 \cdot (-1) \cdot e^{-x}$

$= e^{-x}(3x^2 - x^3)$

$= e^{-x} x^2 (3 - x)$

$f''(x) = (-1)e^{-x}(3x^2 - x^3) + e^{-x}(6x - 3x^2)$

$= e^{-x}(-3x^2 + x^3 + 6x - 3x^2)$

$= e^{-x}(x^3 - 6x^2 + 6x)$

$= e^{-x} \cdot x(x^2 - 6x + 6)$

Stationary points: $f'(x) = 0 \iff \cancel{e^{-x}} x^2(3-x) = 0$

$\Rightarrow x = 0, 3$

$\Rightarrow$ possible extreme points

Inflection points: $f''(x) = 0$ and f'' changes sign

$f''(x) = \cancel{e^{-x}} \cdot x[x^2 - 6x + 6] = 0$

$\Rightarrow x = 0, 3 \pm \sqrt{3}$.

$0 \quad \boxed{3-\sqrt{3}} \quad 3+\sqrt{3}$

$f''(-1) = e^{1} \cdot (-1) \cdot (1 + 6 + 6) > 0$

$f''(1) < 0$

$f(3) > 0$

$f(5) < 0$

$\Rightarrow f''(0)$ changes sign

$\Rightarrow 0$ is an inflection point

$\Rightarrow 3 - \sqrt{3}$ is also an inflection point

$\Rightarrow 3 + \sqrt{3}$ is an inflection point

Tutorial Q3: Extreme points and inflection points

Knowledge used

- Use $f'(x) = 0$ to find stationary points.
- Use $f''(x)$ to classify stationary points and test for inflection points.
- An inflection point requires a sign change in f''.

Solution

$$f'(x) = e^{-x}(3x^2 - x^3) = e^{-x}x^2(3 - x).$$

Hence $f'(x) = 0$ at $x = 0$ and $x = 3$. Also,

$$f''(x) = e^{-x}(6x - 6x^2 + x^3) = e^{-x}x(x^2 - 6x + 6).$$

At the stationary points,

$$f''(0) = 0 \quad \Rightarrow \quad \text{possible inflection point,}$$

$$f''(3) = e^{-3}(18 - 54 + 27) = -\frac{9}{e^3} < 0 \quad \Rightarrow \quad x = 3 \text{ is a local maximum.}$$

Tutorial Q3: Extreme points and inflection points

Knowledge used

- Use $f'(x) = 0$ to find stationary points.
- Use $f''(x)$ to classify stationary points and test for inflection points.
- An inflection point requires a sign change in f''.

Solution

For inflection points, solve $f''(x) = 0$:

$$x = 0 \quad \text{or} \quad x^2 - 6x + 6 = 0.$$

Thus

$$x = 0, \qquad x = 3 - \sqrt{3}, \qquad x = 3 + \sqrt{3}.$$

Since f'' changes sign at each of these points, the inflection points occur at

$$x = 0,\ 3 - \sqrt{3},\ 3 + \sqrt{3}.$$

Tutorial Q4

Question 4

5 mins

By producing and selling Q units of a good, a firm earns total revenue $R(Q) = -0.0016Q^2 + 44Q$ and incurs costs $C(Q) = 0.0004Q^2 + 8Q + 64{,}000$ What production level Q maximises profit? Confirm this is a maximum.

Firm choose Q to max profit.

profit = Revenue − cost. $\pi(Q) = R(Q) - C(Q) = -0.0016Q^2 + 44Q - 0.0004Q^2 - 8Q - 64000$

$= -0.002Q^2 + 36Q - 64000$

$\pi'(Q) = -0.004Q + 36 = 0 \Rightarrow$ stationary point $-0.004Q = -36$

$Q = \frac{36}{0.004} = \frac{36000}{4} = 9000$

First derivative test: $\pi'(Q) > 0$ when $Q < 9000$
$\pi'(Q) < 0$ when $Q > 9000$ $\Rightarrow Q = 9000$ is a global maximum

$\Rightarrow$ max profit

Tutorial Q4: Profit maximization

Knowledge used

- Form the objective function first.
- Use the FOC to find the optimal output.
- Use the second derivative to confirm a maximum.

Solution

$$\Pi(Q) = R(Q) - C(Q) = -0.002Q^2 + 36Q - 64000.$$
$$\Pi'(Q) = -0.004Q + 36 = 0 \Rightarrow Q^* = 9000.$$
$$\Pi''(Q) = -0.004 < 0 \Rightarrow \text{profit is strictly concave.}$$

Hence $Q = 9000$ gives the global maximum, and

$$\Pi(9000) = 98000.$$

Tutorial Q5

Question 5

5 mins

Given the following function:

$$f(x) = \frac{e^x}{2 + e^{2x}}$$

Find the intervals in which the function is increasing and decreasing. Find the minimum and maximum points.

$f'(x)$ $\quad f'(x) \geq 0$ $\quad f'(x) \leq 0$ $\quad$ First derivate test

$f(x) = \frac{e^x}{2+e^{2x}} \Rightarrow$ Quotient rule

$$f'(x) = \frac{e^x(2+e^{2x}) - e^x(2)\cdot e^{2x}}{(2+e^{2x})^2}$$

$$= \frac{e^x(2+e^{2x}-2e^{2x})}{(2+e^{2x})^2}$$

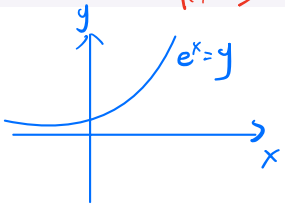
$$= \frac{\boxed{e^x}\boxed{(2-e^{2x})}}{\boxed{(2+e^{2x})^2}}$$

$2 - e^{2x} = 0$

$e^{2x} = 2$

$2x = \ln 2$

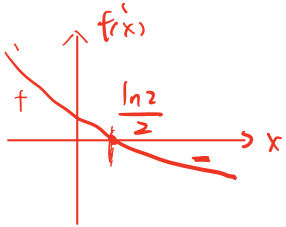
$x = \frac{\ln 2}{2}$

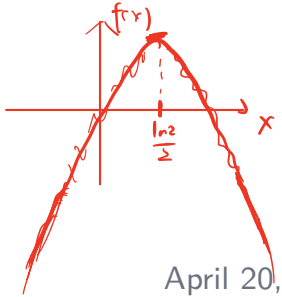


Stationary point: $x = \frac{\ln 2}{2}$

on $(-\infty, \frac{\ln 2}{2}) \Rightarrow f'(x) > 0 \Rightarrow f(x) \nearrow$

on $(\frac{\ln 2}{2}, \infty) \Rightarrow f'(x) < 0 \Rightarrow f(x) \searrow$





First derivative test

$\Rightarrow f(\frac{\ln 2}{2})$ is a global max

No global min

Tutorial Q5: Monotonicity and a global maximum

Knowledge used

- Use the quotient rule to differentiate.
- Use the sign of $f'(x)$ to study monotonicity.
- A sign change from positive to negative gives a maximum.

Solution

$$f'(x) = \frac{e^x(2 - e^{2x})}{(2 + e^{2x})^2}, \qquad f'(x) = 0 \iff x = \frac{\ln 2}{2}.$$

Since $2 - e^{2x} > 0$ for $x < \frac{\ln 2}{2}$ and $2 - e^{2x} < 0$ for $x > \frac{\ln 2}{2}$, f is increasing on $\left(-\infty, \frac{\ln 2}{2}\right)$ and decreasing on $\left(\frac{\ln 2}{2}, \infty\right)$. Therefore

$$x = \frac{\ln 2}{2}$$

is the global maximum point.

Key takeaways

$f(x_1)$

$f'(x_1) \Rightarrow$ rate of change

$x_1 \Rightarrow f(x_1)$

$f(x_1, x_2, x_3)$

$\frac{\partial f(x_1, x_2, x_3)}{\partial x_1} \Rightarrow x_1$ change $\Rightarrow f(x_1, x_2, x_3)$

$\frac{\partial f(x_1, x_2, x_3)}{\partial x_2} \Rightarrow x_2$ change $\Rightarrow f(x_1, x_2, x_3)$

- For global extrema on a closed interval, compare all stationary points and endpoints.
- The first derivative test tracks changes in slope; the second derivative test uses curvature to classify stationary points.
- Inflection points require a sign change in f'', and concavity or convexity often makes optimization much easier.